B. TECH-ME-III-II Sem

23A03604a	ENGINEERING FRACTURE MECHANICS (Professional Elective-II)	L	T	P	C
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Course objectives: The objectives of the course are to	
1	Understanding of Engineering Fracture Mechanics (EFM principles and fatigue crack growth models, with a focus on analyzing and preventing spectacular structural failures.
2	Explore the principles of crack growth and fracture mechanisms, and their applications in material failure analysis.
3	Review the theory of elasticity and explore Westergaard's solution for stress and displacements in Mode I fracture, along with the relationship between the stress intensity factor (K) and the energy release rate (G).
4	Familiarize multi-parameter stress fields for Mode I, Mode II, and mixed-mode fractures, explore the calculation of stress intensity factors (SIF) for various geometries
5	To study fracture toughness testing, crack growth models, analysis, failure assessment diagrams, and mixed-mode fracture, along with methods for crack arrest and repair.

COURSE OUTCOMES On successful completion of this course the student will be able to		
1	Apply LEFM, EPFM, and fatigue crack growth models to assess and prevent catastrophic structural failures at different loads.	*L2,L3,L5*
2	Apply Griffith's theory, calculate energy release rates, and analyze crack propagation mechanisms in materials to predict and prevent fractures in engineering applications.	L3,L4,L5
3	Analyze and apply the theory of elasticity and displacement in Mode I fracture, and understand the connection between the stress intensity factor (K) and the energy release rate (G).	L2,L3,L4
4	Analyze and calculate multi-parameter stress fields for different fracture modes and apply Irwin's and Dugdale's models to understand deformation around crack tips.	L2,L4,L5
5	Perform fracture toughness testing, apply crack growth models, and understand crack closure and failure assessment diagrams.	L2,L3,L5

UNIT 1

EFM Course outline and Spectacular Failures, Introduction to LEFM and EPFM, Fatigue Crack Growth Model

UNIT 2

Crack Growth and Fracture Mechanisms, Griffith TMs Theory of Fracture, Energy Release Rate

UNIT 3

Review of Theory of Elasticity , Westergaard Solution for Stress and Displacements for Mode I, Relationship between K and G

UNIT 4

Introduction to multi parameter stress field for Mode I, Mode II and Mixed Modes, SIF for Various Geometries, Modeling Plastic Deformation, Irwin TMs model, Dugdale Model

UNIT 5

Fracture Toughness Testing, Paris Law and Sigmoidal curve, Crack Closure, Crack Growth Models, J-Integral, Failure Assessment Diagram, Mixed Mode Fracture, Crack Arrest and Repair Methodologies

Text Books

1. Prashant Kumar, Elements of Fracture Mechanics, Tata McGraw Hill, New Delhi, India, 2009.
2. K. R.Y. Simha, Fracture Mechanics for Modern Engineering Design, Universities Press (India) Limited,2001.

Reference Books

1. D. Broek, Elementary Engineering Fracture Mechanics, Kluwer Academic Publishers, Dordrecht, 1986.
2. T.L. Anderson, Fracture Mechanics "Fundamentals and Applications, 3rd Edition, Taylor and Francis Group, 2005.
3. K. Ramesh, e-Book on Engineering Fracture Mechanics, IIT Madras, 2007.

Online Learning Resources:

1. https://nptel.ac.in/courses/112106065
2. https://youtube.com/playlist?list=PLA218B83235A4AD5C&si=XI175OWGIvdMCQH9
3. https://youtube.com/playlist?list=PLA218B83235A4AD5C&si=ruHP1MIsJGNAyMYV
4. https://youtube.com/playlist?list=PLfIFNJ1DPG4ks5AjeCgpbm8nLGM1Pgxer&si=F-fj413KzPAkjPSs